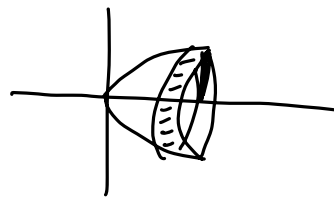


4. $y = 5 \sin(3x) \quad 0 \leq x \leq \frac{\pi}{3} \quad \curvearrowright x\text{-axis}$

$$SA = 2\pi \int (rad) dS$$

$$\left(\frac{dy}{dx}\right)^2 = (15 \cos(3x))^2 = 225 \cos^2(3x)$$



$$dS = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$dS = \sqrt{1 + 225 \cos^2(3x)}$$

$$\text{radius} = y = 5 \sin(3x)$$

$$S.A. = 2\pi \int_0^{\pi/3} 5 \sin(3x) \sqrt{1 + 225 \cos^2(3x)} dx$$

$$\begin{aligned} u &= \cos(3x) \\ du &= -3 \sin(3x) dx \end{aligned}$$

$$-\frac{1}{3} du = \sin(3x) dx$$

$$\text{Top: } u = \cos(\pi)$$

$$\text{bot: } u = \cos(0)$$

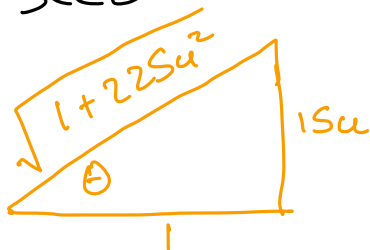
Trig sub

$$15u = \tan \theta$$

$$15 du = \sec^2 \theta d\theta$$

$$\sqrt{1 + \tan^2 \theta}$$

$$= \sec \theta$$



$$= 2\pi \int \cancel{5 \sin(3x)} \sqrt{1 + 225 u^2} \left(-\frac{1}{3} \cancel{\sin 3x} \right) dx$$

$$= -\frac{10\pi}{3} \int_1^{-1} \sqrt{1 + 225 u^2} du = \frac{10\pi}{3} \int_{-1}^1 \sqrt{1 + 225 u^2} du$$

$$= \frac{10\pi}{3} \int \frac{\sec \theta \sec^2 \theta d\theta}{15} = \frac{2\pi}{9} \int \sec^3 \theta d\theta$$

$$= \frac{2\pi}{9} \cdot \frac{1}{2} \left[\frac{\sec \theta \tan \theta}{\sqrt{1+225u^2} \cdot 15u} + \ln \left| \frac{\sec \theta + \tan \theta}{\sqrt{1+225u^2} \cdot 15u} \right| \right]_{u=-1}^{u=1}$$

$$= \frac{\pi}{9} \left[\sqrt{1 + 225 u^2} \cdot 15u + \ln \left(\sqrt{1 + 225 u^2} + 15u \right) \right]_{-1}^{+1}$$

$$= \frac{\pi}{9} \left[\sqrt{226} \cdot 15 + \ln(\sqrt{226} + 15) - \sqrt{226}(-15) - \ln(\sqrt{226} - 15) \right]$$

$$= \frac{\pi}{9} \left[30\sqrt{226} + \ln(\sqrt{226} + 15) - \ln(\sqrt{226} - 15) \right]$$

$$\approx 159.8$$

$$y = 2x^2 + 4$$

$$x=0 \text{ to } x=2$$

$$\int \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\left(\frac{dy}{dx}\right)^2 = (4x)^2 = 16x^2$$

$$\int_0^2 \sqrt{1 + 16x^2} dx$$

2.0.2 Example:

Remove the parameter from the parametric equations

$$x = \ln(2t), y = 2t + 5$$

$$y - 5 = 2t$$

$$\frac{y-5}{2} = t$$

$$x = \ln\left(2\left(\frac{y-5}{2}\right)\right)$$

$$x = \ln(y-5)$$

2.0.3 Example:

Remove the parameter from the parametric equations

$$x = e^t - 1, y = e^{3t}$$

$$x+1 = e^t$$

$$y = e^{3t}$$

$$y = (e^t)^3$$

$$y = (x+1)^3$$

10.2 Parametric Calculus

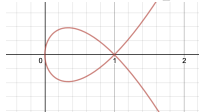
1 The Motion of a Function / Particle

It's helpful to think of parametric equations as describing the position of a particle. The main idea behind parametric calculus is that the motions of x and y don't depend (directly) on each other, but depend on time. Thus we are interested in the derivatives $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

derivative	positive	negative	zero
$\frac{dx}{dt}$	right	left	stopped horizontally
$\frac{dy}{dt}$	up	down	stopped vertically

1.0.1 Example:

Consider a particle whose position is given by the equations $x = t^2$, $y = t^3 - t$.



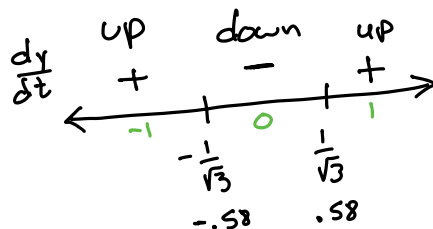
When is the particle moving up/down?

$$\frac{dy}{dt} = 3t^2 - 1 = 0$$

$$3t^2 = 1$$

$$t^2 = \frac{1}{3}$$

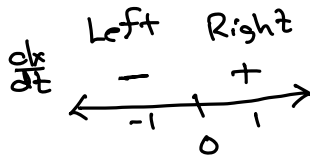
$$t = \pm \frac{1}{\sqrt{3}}$$



When is it moving left/right?

$$\frac{dx}{dt} = 2t = 0$$

$$t = 0$$



Left $(-\infty, 0)$
Right $(0, \infty)$

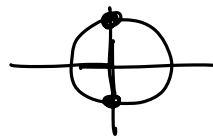
1.1 Tangents

Notice that the curve in the previous example had both horizontal and vertical tangents.

Vertical tangents can occur when $\frac{dx}{dt} = 0$ Horizontal tangents can occur when $\frac{dy}{dt} = 0$

1.1.1 Example:

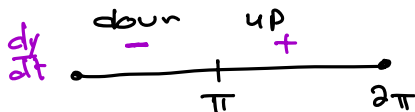
Consider a particle whose position is given by the equations $x = \sin(2t)$, $y = \cos(t)$ for $0 \leq t \leq 2\pi$. Determine when the particle is moving up, down, left, and right, and find the values of t where the graph has horizontal and vertical tangents.



Vertical

$$\frac{dy}{dt} = -\sin t = 0$$

$$t = 0, \pi$$



down: $0 \leq t \leq \pi$

up: $\pi \leq t \leq 2\pi$

Horiz

$$\frac{dx}{dt} = 2\cos(2t) = 0$$

$$\cos(2t) = 0$$

$$2t = \frac{\pi}{2} + 2\pi n$$

$$2t = \frac{3\pi}{2} + 2\pi n$$

$$t = \frac{\pi}{4} + \pi n$$

$$t = \frac{3\pi}{4} + \pi n$$

$\frac{\pi}{4}, \frac{5\pi}{4}$	$\frac{3\pi}{4}, \frac{7\pi}{4}$
.78, 3.92	2.35, 5.5

1.2 Equations of Tangent Lines

Suppose you want to find the equation of a tangent line (in $y = mx + b$ form) at a given value of t . We need to know that

$$\text{slope} = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

1.2.1 Example:

Find the equation of the line tangent to the curve $x = t^2 - 2t$, $y = \sqrt{t}$ at the point where $t = 4$.

Find Slope $\frac{dy}{dx}$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t}}$$

at $t=4$

$$\Rightarrow \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$\frac{dy}{dx} = \frac{\frac{1}{4}}{\frac{1}{6}} = \frac{1}{24}$$

$$\frac{dx}{dt} = 2t - 2 \Rightarrow 2(4) - 2 = 6$$

Find the point

$$x = 4^2 - 2(4) = 8 \quad (8, 2)$$

$$y = \sqrt{4} = 2$$

Plug into $y - y_1 = m(x - x_1)$

$$\boxed{y - 2 = \frac{1}{24}(x - 8)}$$

2 Parametric Arc Length

We now use the parametric form of dS :

$$dS = \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2} dt$$

We find arc length by using the formula

$$\text{distance traveled} = \int_{\alpha}^{\beta} dS.$$

If we want the length of the curve, we must ensure that **the curve is only traversed once** during the interval $t = \alpha$ to $t = \beta$. The next two examples show the difference in this:

2.0.1 Example:

Find the arc length, and the distance traveled by a particle with these parametric equations:

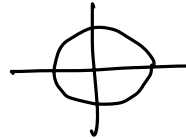
$$x = \sin t, \quad y = \cos t, \quad 0 \leq t \leq 2\pi$$

$$\left(\frac{dx}{dt}\right)^2 = (\cos t)^2 = \cos^2 t \quad \left(\frac{dy}{dt}\right)^2 = (-\sin t)^2 = \sin^2 t$$

$$dS = \sqrt{\cos^2 t + \sin^2 t} dt = 1 dt$$

$$\text{Arc length} = \int_0^{2\pi} dt = t \Big|_0^{2\pi} = \boxed{2\pi}$$

↑
Arc length & distance traveled



2.0.2 Example:

Find the arc length, and the distance traveled by a particle with these parametric equations:

$$x = \sin(3t), \quad y = \cos(3t), \quad 0 \leq t \leq 2\pi$$

$$\left(\frac{dx}{dt}\right)^2 = (3\cos(3t))^2 = 9\cos^2(3t) \quad \left(\frac{dy}{dt}\right)^2 = (-3\sin(3t))^2 = 9\sin^2(3t)$$

$$\underline{\text{Dist} = 6\pi} \quad \text{Arc length} = 2\pi$$

$$dS = \sqrt{9\cos^2(3t) + 9\sin^2(3t)} dt = \sqrt{9(\cos^2 3t + \sin^2 3t)} dt = \sqrt{9} dt = 3 dt$$

$$\text{Dist} = \int_0^{2\pi} 3 dt = 3t \Big|_0^{2\pi} = 3(2\pi) = \boxed{6\pi}$$

2.0.3 Example:

Find the length of the curve $x = e^t + e^{-t}$, $y = 5 - 2t$, $0 \leq t \leq 3$

$$\left(\frac{dx}{dt}\right)^2 = (e^t - e^{-t})^2 = \left(\frac{dy}{dt}\right)^2 = (-2)^2 = 4$$

$$= e^{2t} - 2e^t e^{-t} + e^{-2t}$$

$$= e^{2t} - 2 + e^{-2t}$$

$$dS = \sqrt{e^{2t} - 2 + e^{-2t} + 4} dt = \sqrt{e^{2t} + 2 + e^{-2t}} dt$$

$$= \sqrt{(e^t + e^{-t})^2} dt = (e^t + e^{-t}) dt$$

$$\int_0^3 e^t + e^{-t} dt = \left[e^t - e^{-t} \right]_0^3 = \left(e^3 - \frac{1}{e^3} \right) - (1 - 1)$$
$$= \boxed{e^3 - \frac{1}{e^3}}$$

3 Parametric Surface Area of Revolution

Recall: Surface area $= 2\pi \int_{\alpha}^{\beta} (\text{radius}) dS$

Here we use the parametric dS , and we have to be sure that it's only traversed once.

3.0.1 Example:

The curve $x = 2 \sin t$, $y = 3 \cos t$, $0 \leq t \leq \pi$ is rotated around the y axis. Find the surface area.